

Government Engineering College Kishanganj

Department of Computer Science and Engineering

COMPUTER NETWORKS LAB-02

Exploring Packet Sniffer and Packet Analyzer

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Experiment	LAB-02
Tool Used	Wireshark
Capture File	LAB_CN_02_NEW.pcap

1. Protocols observed at different layers

Layer	Observed protocols / evidence
Application	HTTP, TLS 1.2, TLS 1.3, QUIC, and SSDP. HTTP GET traffic and TLS Client Hello messages are visible.
Transport	TCP and UDP. HTTP/neverssl uses TCP port 80; HTTPS/TLS conversations use TCP port 443; QUIC uses UDP.
Network	IPv4 is clearly visible, including 10.85.19.11 communicating with 34.223.124.45.

2. Total amount of data received

Case	Result
A. http://neverssl.com/	About 5 KB received in the server → client direction for TCP stream 57. The HTTP response advertises Content-Length: 3961 bytes.
B. https://www.cornell.edu/	The supplied screenshots do not show the aggregate received-byte total for the Cornell transaction.

3. Analysis of http://neverssl.com/

A. HTTP GET request

Item	Observed result
Number of HTTP GET requests	1 HTTP GET request is visible for the neverssl connection in TCP stream 57.
GET request	GET / HTTP/1.1 Host: neverssl.com User-Agent: curl/8.21.0 Accept: */*

B. HTTP version used by the server

Question	Answer
HTTP version	The server response is HTTP/1.1 200 OK; therefore, the server is using HTTP/1.1 for this request.

C. TCP segments and response data

GET request	Server → client TCP segments	HTTP response data
GET / HTTP/1.1	6 segments	Content-Length = 3961 bytes

Evidence / interpretation
The Conversations window shows 6 server → client packets and about 5 KB on the wire. The HTTP response itself advertises 3961 bytes of content.

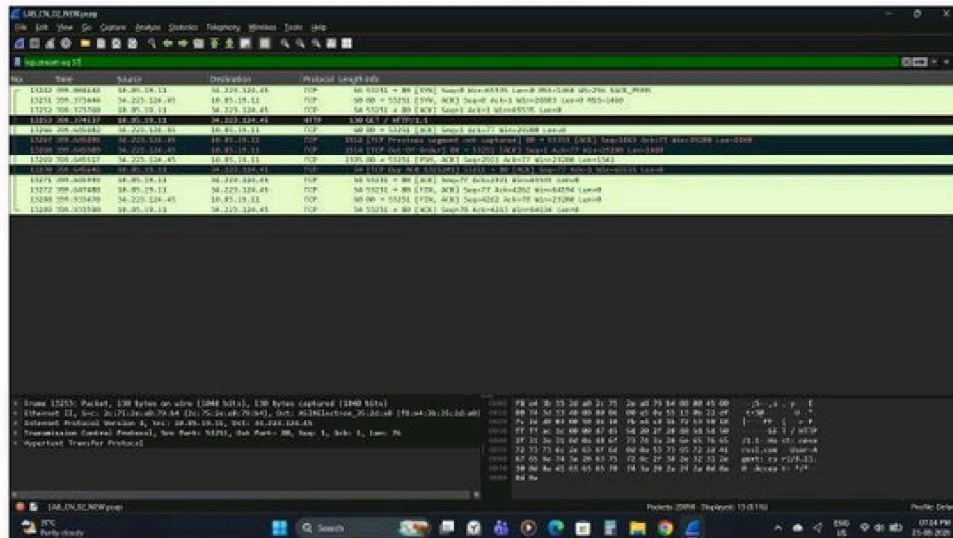
D. TCP three-way handshake

Step	Packet	Sequence	Acknowledgment
1	Client → Server: SYN	0	—
2	Server → Client: SYN-ACK	0	1
3	Client → Server: ACK	1	1

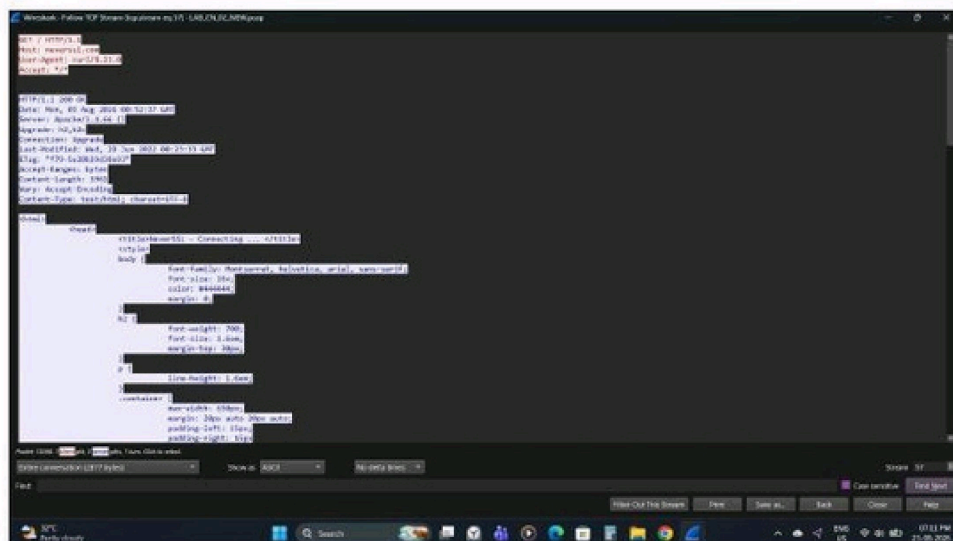
Connection	10.85.19.11:53251 → 34.223.124.45:80. The packet list shows SYN, SYN-ACK and ACK in that order.
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Wireshark Evidence

Selected supplied screenshots used as evidence for the report.

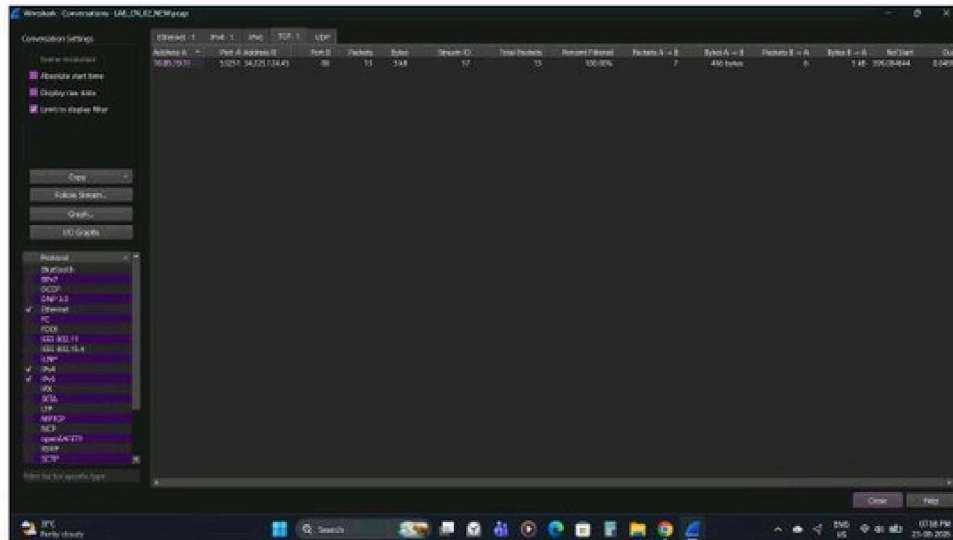


TCP stream 57 and neverssl HTTP/TCP packet details.

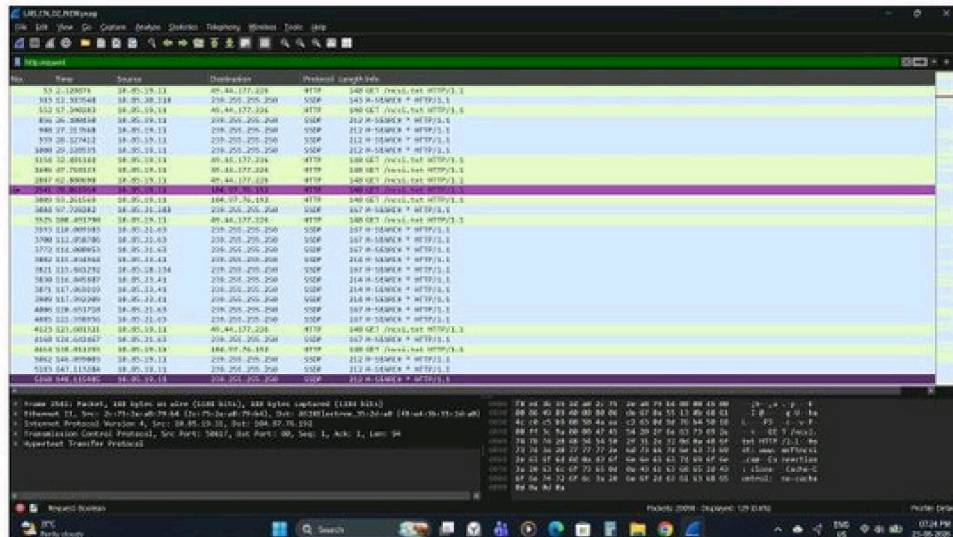


Follow TCP Stream: HTTP/1.1 response and Content-Length 3961.

Wireshark Evidence

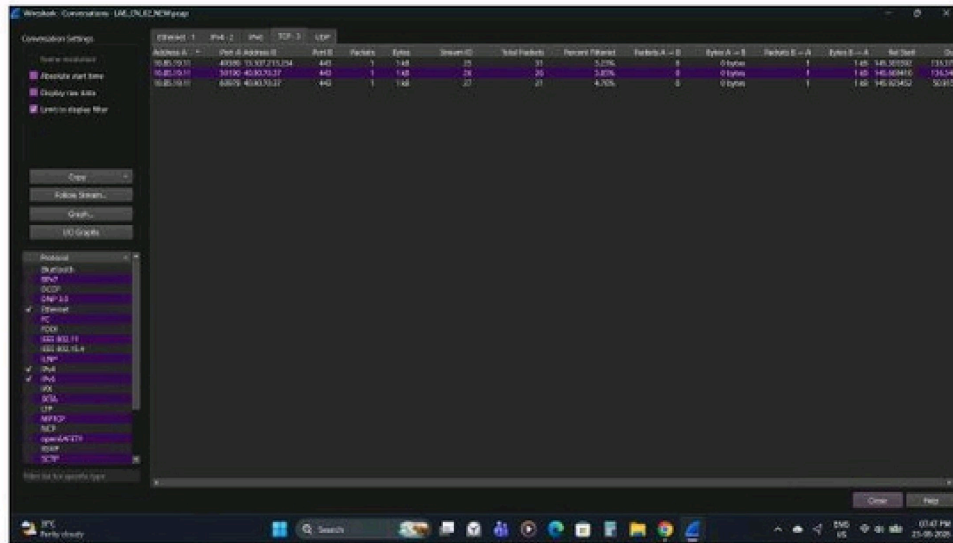


Conversations: stream 57 and approximately 5 KB server-to-client.



TLS Client Hello/SNI traffic in the capture.

Wireshark Evidence



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.0.2.15	10.0.2.15	HTTP	100	GET / HTTP/1.1
2	0.000000	10.0.2.15	10.0.2.15	TCP	60	40000 → 34100 [RST] Seq=40000 Win=0 Len=0
3	0.000000	10.0.2.15	10.0.2.15	TCP	60	34100 → 40000 [ACK] Seq=34100 Win=0 Len=0

HTTPS/TCP conversations from the supplied capture.